

PRODUCT REQUIREMENT DOCUMENT

EasyReads

A Mindful Mobile Ecosystem Built to Foster Lifelong Reading Habits for Beginners Without Friction or Failure.

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Target Stack: React Native + Firebase Backend

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1. Executive Vision & Philosophy

The primary reason beginner readers fail to maintain a long-term reading habit is not a lack of interest, but the heavy cognitive load and psychological friction introduced by traditional tracking methods. Most existing apps act as aggressive performance trackers, counting streaks rigidly, serving complex charts, and inducing "comparison anxiety." This translates an organic, relaxing hobby into unneeded homework or chores.

The "Anti-Homework" Core Philosophy

EasyReads is intentionally designed as an antidote to performance tracking. It minimizes gamification anxiety while maintaining structural encouragement. The environment focuses entirely on psychological safety: a forgiving streak model, localized context dictionary popups, and the isolation of analytics to personal progress rather than social leaderboards.

Core Pillars of the App

- **Zero Commercial Friction:** 100% free with no paywalls, subscriptions, premium tiers, or intrusive ad placements.
- **Micro-Progress Metrics:** Focusing purely on natural metrics: pages read and individual words discovered.
- **Forgiving Accountability:** Adapting milestones to fit the realities of unpredictable human schedules through intelligent algorithms.

2. Technical Architecture & Data Model

The technical blueprint relies entirely on a modern, decoupled, lightweight framework optimized for rapid iteration, offline execution capabilities, and zero-maintenance cost.

The Tech Stack

Layer	Technology Selected	Strategic Implementation Rationales
Frontend	React Native (Expo framework)	Allows complete native platform rendering across Android and iOS via a single codebase, drastically decreasing initial development time.
Database & Auth	Firebase (Firestore & Auth)	Real-time synchronization out-of-the-box, reliable native caching layer for offline storage support, and painless OAuth integration.
Dictionary Service	Free Dictionary API (REST)	Allows immediate, zero-authentication JSON lookup requests for definitions and audio pronunciation streams.

Core Firestore Document Schemas

The following structural document modeling defines the underlying Firebase collection layout:

Users Collection (/users/{uid})

```
{
  "uid": "STRING (Primary Key)",
  "displayName": "STRING",
  "email": "STRING",
  "createdAt": "TIMESTAMP",
  "currentStreak": "INTEGER",
  "streakFreezeAvailable": "INTEGER (Max 2)",
  "rollingPageAverage": "FLOAT"
}
```

```
{
  "bookId": "STRING",
  "title": "STRING",
  "author": "STRING",
  "totalPages": "INTEGER",
  "pagesRead": "INTEGER",
  "status": "STRING [reading | completed]",
  "startedAt": "TIMESTAMP",
  "estimatedFinishDate": "TIMESTAMP"
}
```

3. Detailed Core Features & Functional Specs

3.1 The "Currently Reading" Dynamic Card

Positioned as the centerpiece of the home interface, this card eliminates layout noise by presenting exactly one single active focal point. It removes multi-tier menu hopping and instantly prompts action.

- **Progress Indicator:** A sleek visual loading/progress bar translating raw tracking counts into a percentage format.
- **Dynamic Finish Predictor:** Calculates the expected finish date using a basic rolling arithmetic mechanism. Let total remaining pages be P_{rem} and the user's rolling daily average speed over the past 5 days be A_{page} . The remaining days D_{rem} is computed as:

$$D_{rem} = \lceil P_{rem} / A_{page} \rceil$$

3.2 The Smart Goal System

Traditional trackers require users to manually declare rigid goals (e.g., "Read 30 pages daily"). When an active schedule compromises a day's reading, the user faces guilt and abandons the app. EasyReads dynamically absorbs reading behavior and self-adjusts.

DYNAMIC ADJUSTER ALGORITHM

$$\text{Goal} = \text{Baseline} \pm (\text{Deviation} * 0.2)$$

If a user struggles continuously across three consecutive days, the engine recalibrates downward to lower barriers, ensuring the user experiences achievable milestones and feels empowered to build momentum again.

| 3.3 Forgiving Streak Engine (Streak Freeze)

To completely clear the negative psychological impact of breaking habits, EasyReads introduces an implicit "Streak Freeze" system.

- Every account receives a maximum replenishment cap of 2 Streak Freezes per week.
- If a calendar date logs zero pages read, the engine scans the user data. If a freeze token is available, it is spent automatically.
- The system alerts the user with a comforting, warm notification: *"You had a busy day yesterday! We protected your streak for you. Relax and pick it back up today."*

| 3.4 Context Vocabulary Lookup & Social Sharing Cards

While executing a reading session, users naturally stumble upon unfamiliar language. Exiting an app to check Google creates immediate drop-off loops.

1. The user quickly inserts or highlights an unfamiliar term within a global search overlay.
2. The application performs an immediate asynchronous background lookup to the open dictionary endpoint.
3. The return JSON maps straight into a simple modal container displaying: **[Word] | [Simple Definition] | [Audio Playback Icon]**.
4. **The Vocab Card Feature:** Users can tap a single action item to export this word into a beautiful, minimalistic graphic card containing the word and definition, ready for easy social sharing without displaying any underlying app competitive metrics.

| 3.5 Milestone Celebration: "Pehla Kitaab"

Finishing a first book is a major transitional phase for non-readers. EasyReads locks this specific event into a dedicated experiential interaction design.

When the local updates state transitions a book status field to completed for the very first time, all basic interfaces lock out to launch a delightful fullscreen event canvas. This presents beautiful cultural motifs, subtle sound cues, and a certificate layout congratulating them on completing their first milestone, avoiding numerical point scoring or performance tiers.

4. Interface Design & Visual System

To communicate calmness and clarity, the layout avoids sharp, neon, contrasting color schemes, choosing muted tones, organic rounded corners, and extensive negative space instead.

Color Archetype Palette

Color Element	Hex Value	Design Application
Primary Canvas Background	#FAF8F5	Warm, eye-strain reducing soft cream tone mirroring physical book paper texture.
Brand Accent (Deep Ink)	#1A2E40	High-contrast, elegant deep blue-black used for headers and foundational typography.
Calming Secondary Tone	#4A7C59	Muted organic sage green applied to active trackers, progress metrics, and button UI elements.
Soft Borders & Dividers	#E2E8F0	Light gray dividers providing clear structural containment without introducing visual clutter.

Wireframe Interface Flows

Screen 1: The Focused Dashboard

```
+-----+
| [Logo: EasyReads]                                (Profile) |
+-----+
|
| Namaste, Rishabh!
| Your daily smart goal is: 12 Pages Today
|
| +-----+
| | CURRENTLY READING |
| |                   |
| | "Atomic Habits"   |
| | By James Clear    |
| |                   |
| | [===== PROGRESS: 45% =====] |
| | 90 / 200 Pages Read |
| |                   |
| | Estimated Completion: July 12, 2026 |
| +-----+
|
| [ + Update Progress ]      [ 🔍 Quick Word Lookup ]
|
| +-----+
| | 🔥 5-Day Habit Streak      [ ❄️ 2 Freezes Available] |
| +-----+
+-----+
```

Screen 2: The Vocabulary Sheet Modal

```
+-----+
|
|   <- Back
|
+-----+
| | WORD DICTIONARY SEARCH | |
| | [ Resilient           ] [ Search ] | |
| +-----+
|
| Result Found:
| re·sil·ient (/rə'zilyənt/) [🔊 Listen]
| adj. Able to withstand or recover quickly from
| difficult conditions.
|
| +-----+
| | Save to My Vocabulary Notebook | |
| +-----+
| | ✨ Generate & Share Visual Vocab Card | |
| +-----+
|
+-----+
```

5. Engineering Execution Blueprint

To convert this document into tangible progress, follow this phased milestone schedule for your engineering sprints:

Milestone 1: Foundations & Backend Pipeline (Weeks 1-2)

- Initialize an Expo React Native environment using TypeScript template structures.
- Configure Firebase Core SDK integrations; establish standard baseline Firestore security rules.
- Develop the complete email/password and anonymous authentication pipeline natively inside the frontend state.

Milestone 2: Database Crud & Smart Track Engine (Weeks 3-4)

- Build the active client UI forms allowing users to create new books in the system.

- Write the mathematical recalculation functions inside React state hooks to dynamically update progress tracking metrics.
- Implement the automated local date check systems to calculate streak data and handle automatic streak freeze deduplication.

Milestone 3: API Integration & Fine Polishing (Weeks 5-6)

- Connect Axios/Fetch pipelines with the external dictionary endpoints to manage quick-search inputs seamlessly.
- Incorporate design systems across components, standardizing fonts, margins, and color presets globally.
- Write the operational conditions monitoring the completion state of an user's first book, ready to safely trigger the full-screen confetti animation canvas.